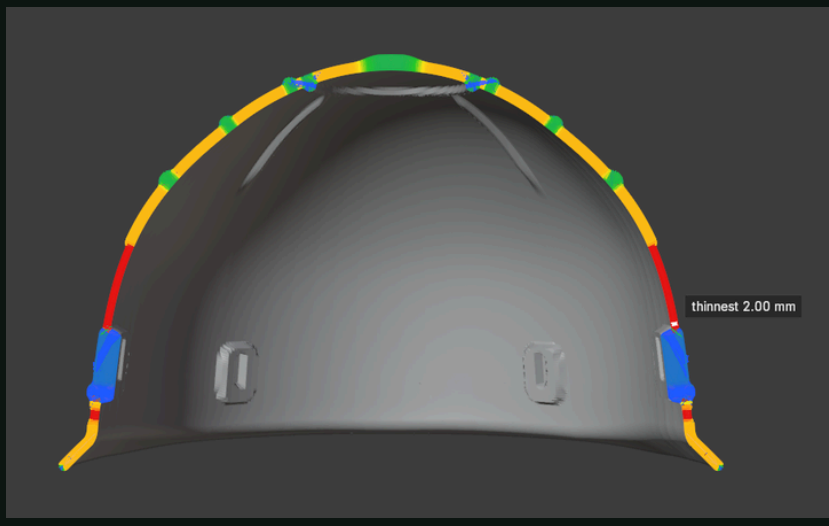


Look under the skin. Find the thin wall.

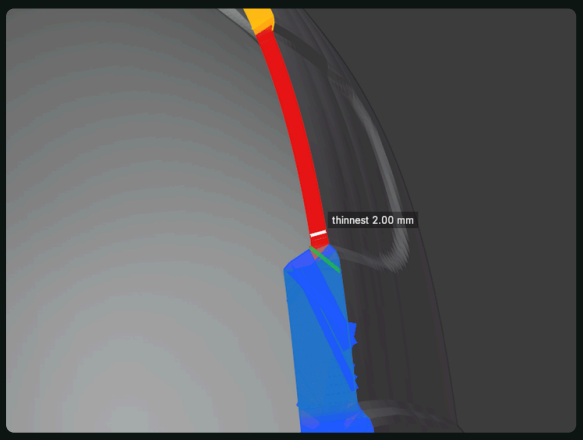
Reinforce it if needed.

Three tools in one panel (a visual guide on a closed, repaired mesh, not a simulation). **1 · Cross Section**: slide and tilt a plane through the part and see the wall thickness of the cut in color, with histograms. **2 · Analysis**: measure every face, select the thin ones, and the weight. **3 · Reinforce**: add meat, ribs, a grid or a honeycomb where it is thin, always on a copy.

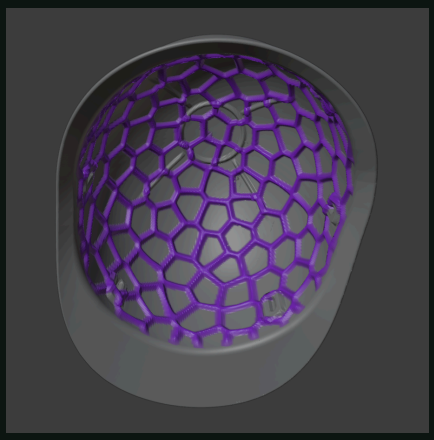
1 · CROSS SECTION | 2 · ANALYSIS | 3 · REINFORCE

ANY ANGLE | WEIGHT · 28 MATERIALS

SECTION HEATMAP · THINNEST SPOT



EXTRA · HONEYCOMB, VIEW FROM INSIDE



3 tools
cross section, analysis, reinforcement, one panel with tabs

≈ 50 ms
per plane move on a 2.9 M-triangle scan, so it follows the mouse

1 % under
a realistic minimum next to the single thinnest spot

Safe
the cut is a modifier, reinforcement goes on a copy; Esc cancels

Blender 4.2+ · Windows · No extra packages · GPL-3.0-or-later

Cover: 532k-triangle helmet standing level, cut across Y, Min 2.5 mm / OK 4 mm

01 GET STARTED

Install it. Select the mesh. Plane, cut, heatmap.

Visual guide only

MeshUnderskin is not a metrology tool and gives no certified or guaranteed results. It contains no simulation and no strength, fatigue or FEA calculations. Colours and numbers only indicate wall thickness measured on the mesh. Always check the real part with proper measuring equipment and testing.

Install

Edit › Preferences › Get Extensions › ⌄ › **Install from Disk** › **MeshUnderskin_1.6.3_Gumroad.zip** (do not unzip), then **N** › **MeshUnderskin** in the 3D Viewport.

Requirements

Windows (developed and tested on Windows, Blender 5.2.3 LTS; Linux and macOS are not tested). Blender 4.2 LTS or newer. No extra Python packages.

Units

Every length is shown in the scene's length unit (Scene › Units). Set **Millimeters** for parts.

License

GPL-3.0-or-later, full source. © 2026 Mirosław Pałucki.

First set the limits of your part

Min thickness: anything thinner is red. **OK thickness**: from here it is green. Twice OK is blue: there you could save material. In between it goes orange → yellow → green. Both values are used by the section heatmap and by the whole-object analysis. Prefer a plain scale? Switch **Colors** to **Range** and set it from–to in mm (page 06).

COLOR	MEANING
red	thinner than Min: add material
orange · yellow	between Min and OK: marginal
green	OK up to 2 × OK
blue	2 × OK or more: thick
grey	no opposite wall found

STEP 1 · PLANE

Add Section Plane

A plane sized to the object (its size in the plane + 12 %). Nothing is cut yet, so even a dense scan stays quick while you place it.

1. Pick the axis **X · Y · Z**, tilt it if needed (**Tilt**).
2. **Move with Mouse**, the **– / +** buttons, the slider or a typed coordinate.

STEP 2 · CUT

Cut Object

A Boolean modifier removes one side of the plane. The mesh is not edited: **Uncut** shows the whole object again.

1. **Flip side** keeps the other half.
2. **Color section cap** paints the cut face red; **Solver**: Manifold, Exact or Float.

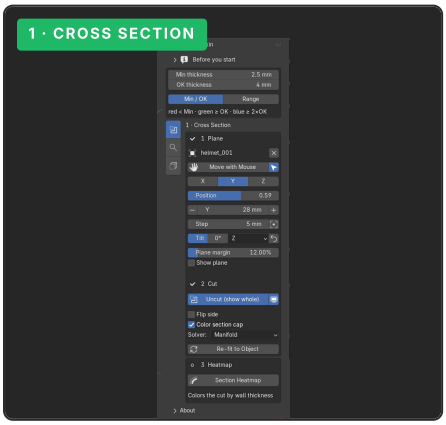
STEP 3 · HEATMAP

Section Heatmap

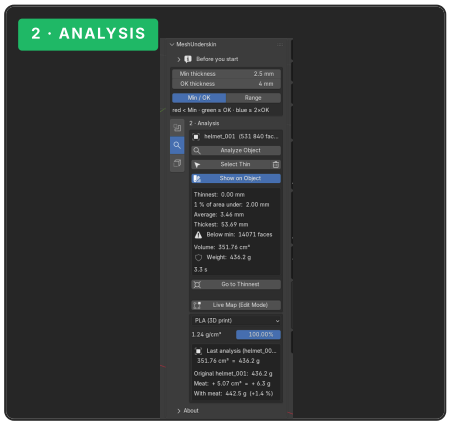
The cut is colored by wall thickness and follows the plane live. Works with or without the cut.

1. **Histogram** and **Mark thinnest** in the viewport.
2. **Go to Thinnest** puts the 3D cursor on it.

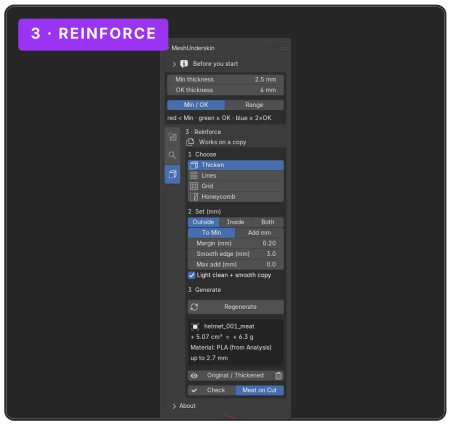
One panel, three tabs



Plane, cut, heatmap and histograms.



Whole object, thin faces, weight.



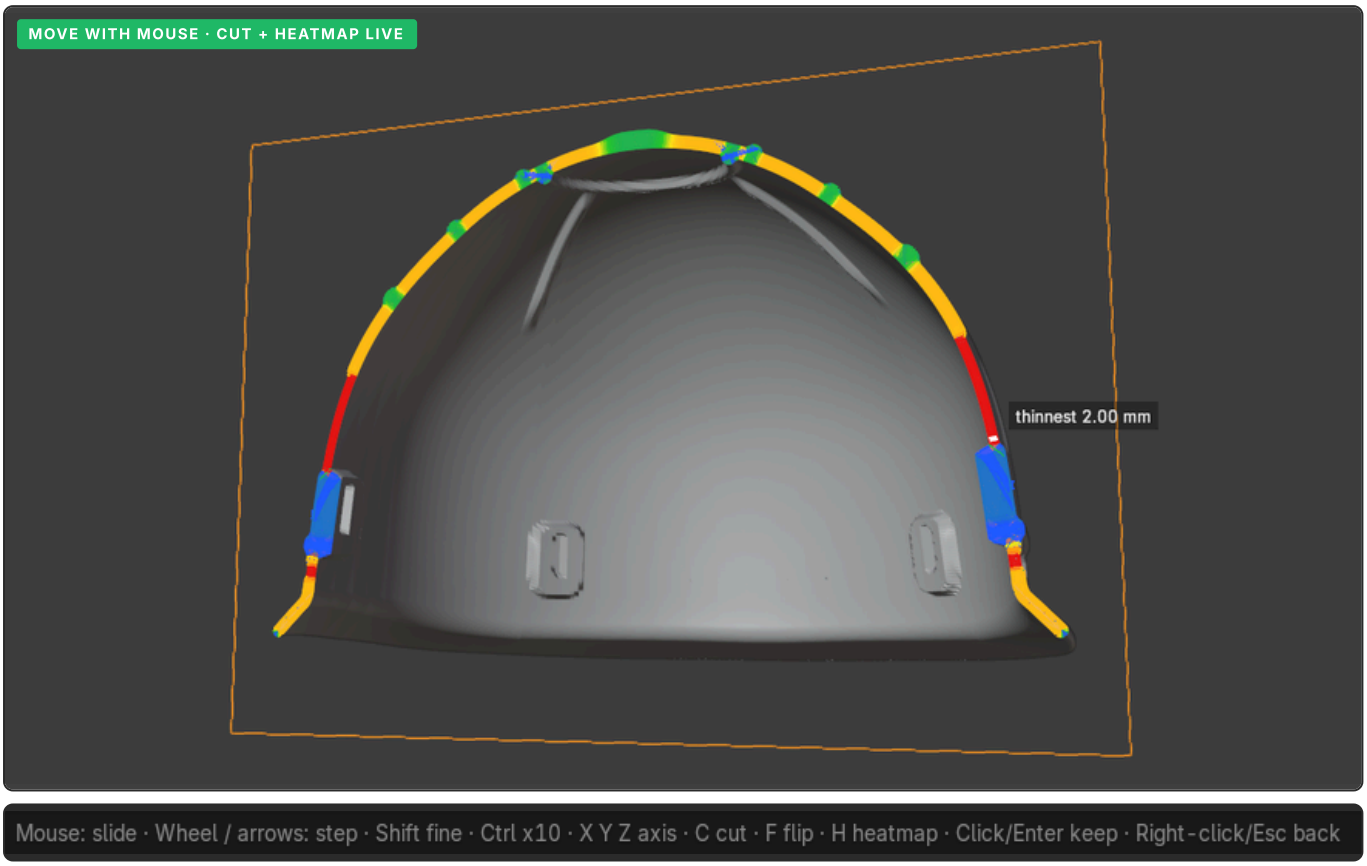
Add meat or ribs, always on a copy.

Vertical tabs on the left, like ThreadPharmacy. **Min**, **OK** and the colors sit on top and are shared by all tabs.

02 MOVE THE PLANE

Slide it with the mouse. Step it. Tilt it.

Press **Move with Mouse** and the plane follows the mouse along its axis: 80 % of the view width is the whole object. The wheel and the arrow keys move it one **Step** (about 1 % of the part, a round number in your unit). The header shows the position, the status bar shows the keys.



Status bar while moving. The heatmap and the histogram update on every move, about 50 ms on a 2.9 M-triangle scan.

Keys while moving

KEY	DOES
`mouse`	slide the plane along the axis
`wheel` `← →` `↑ ↓`	one step back / forward
`Shift`	fine: one tenth of the move or step
`Ctrl` + wheel	ten steps
`X` `Y` `Z`	change the axis (the plane is fitted again)
`[` `]`	tilt the plane −5° / +5° (Shift: 1°)
`C` · `F` · `H`	cut on / off · flip side · heatmap on / off
`click` `Enter`	keep the new position
`right-click` `Esc`	back to where you started (axis, tilt, side, cut)
`middle mouse`	orbit the view as usual while moving

Other ways

- **− / +** next to the coordinate: one step, one undo step each.
- **Coordinate** field: type the exact position, e.g. `-12.5`.
- **Position** slider: 0 = one end of the part, 1 = the other.
- **Center** puts the plane in the middle.
- **Tilt** slider (−89° … +89°) turns the plane around one of the two other axes (**Around**); the arrow icon sets it back to 0°. The cut, the heatmap and the histogram follow.
- **Grab** (arrow icon) selects the plane, so `G` and the gizmo move the plane, not your object.
- **Re-fit to Object** after you changed the mesh: same position, same cut.

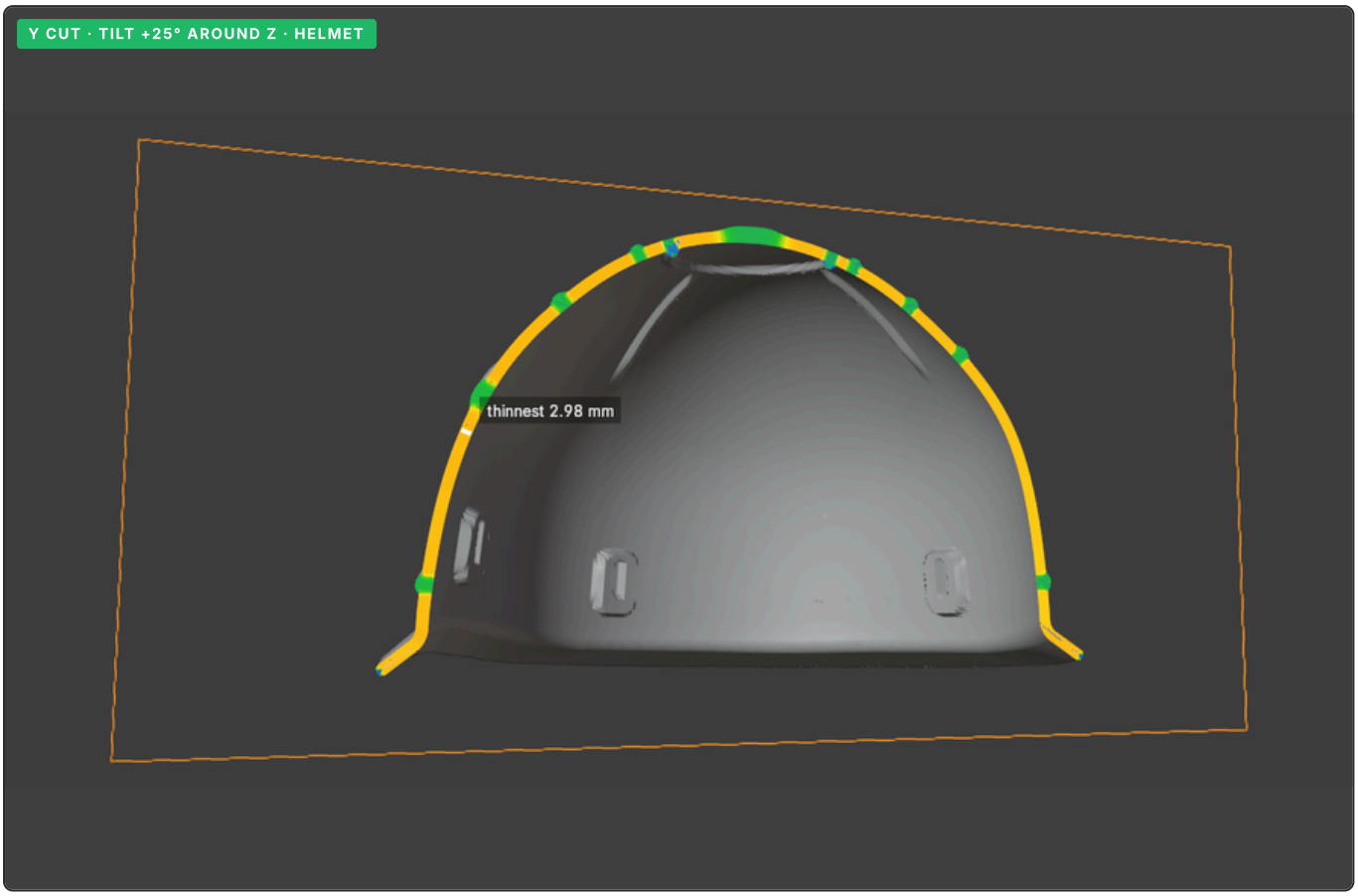
Plane first, cut later

Positioning without the cut is instant even on big scans. Cut when the plane is where you want it, or leave it uncut and read the heatmap only.

03 ANY ANGLE

Tilt the plane until it is square to the wall.

A wall cut at a slant looks thicker than it is. Tilt the plane until it cuts the wall at a right angle and the heatmap shows the true thickness. **Tilt** goes from −89° to +89° around one of the two other axes (**Around**); while moving with the mouse, `[` `]` tilt by 5° (Shift: 1°).



The same 532k-face helmet, cut across Y and tilted 25° around Z. The plane, the cut, the heatmap and the histogram all follow; the histogram title shows the angle. Tilting lets you cross walls that a straight cut runs along.

Same tools at any angle

Cut / Uncut, Flip, the three histograms, opacity, Go to Thinnest, the edge graph and the purple meat band all work on a tilted plane.

Plane still fits

The tilted plane is sized again so it still covers the whole part (+ margin).

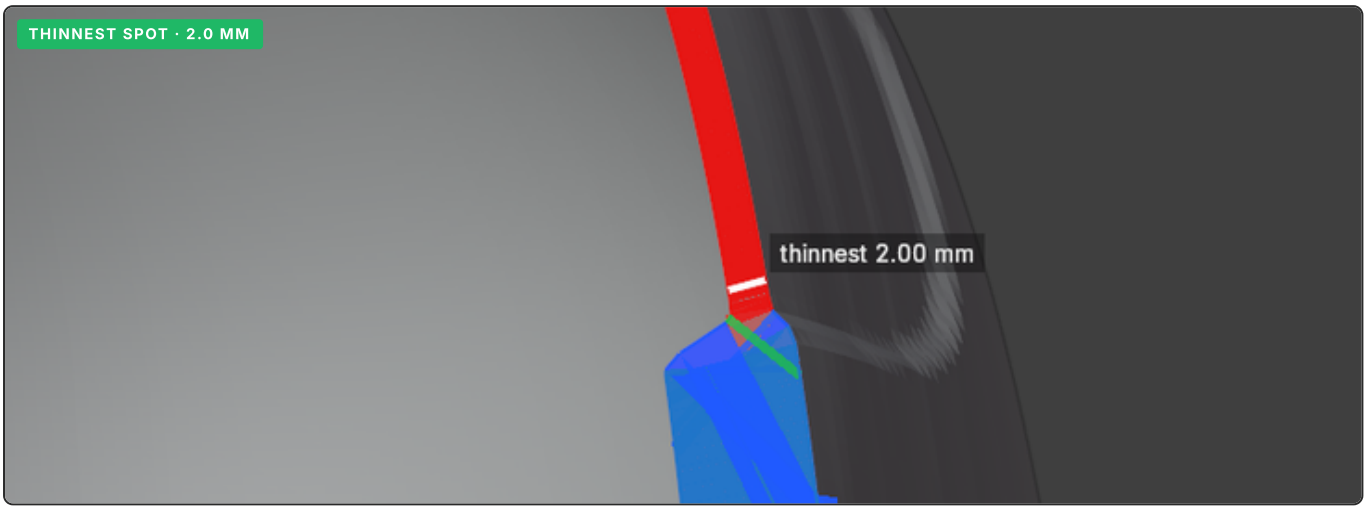
Speed

Straight cuts use a fast pre-filter per axis. A tilted cut tests every triangle, so on multi-million-face scans a move takes a little longer.

04 SECTION HEATMAP

The cut tells you where the wall is thin.

From every point of the cut contour a ray goes into the wall, in the cut plane, along the surface normal, until it leaves the material at the opposite wall. That distance is what a caliper would show on the real cut part. The cut face is filled with that color, wall by wall.



Mark thinnest draws a white line across the thinnest wall of this cut and labels it. **Go to Thinnest** puts the 3D cursor there and centers the view.



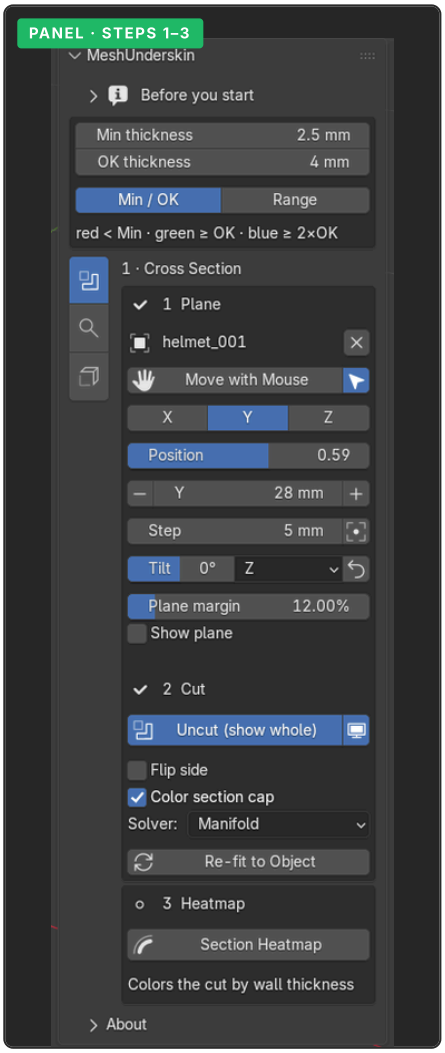
Histogram, bottom left of the viewport: above it thinnest, **1 % under**, average, thickest and how much of the cut is below Min; below it the color scale. Three styles: page 06.

Read the numbers

VALUE	MEANS
Thinnest	one single spot. Scan noise or a sharp rim can make it tiny.
1 % of the cut under	only 1 % of the cut is thinner: use it as the realistic minimum.
Average · Thickest	weighted by contour length.
Below min	share of the cut that needs more material.

A wall cut at an angle

The heatmap shows the thickness in the cut plane, like a real cut. A wall that runs at a slant to the plane looks thicker here; tilt the plane until it is square to the wall, or use **Whole Object** for the true normal thickness.

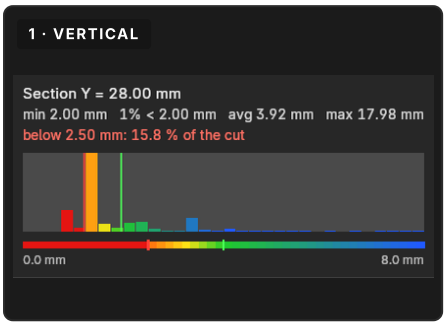


On Top draws the heatmap over everything (default; always on when not cut).

05 COLORS · HISTOGRAMS · OPACITY

Your scale, three histograms, and see-through walls.

Choose how thickness becomes color, how the distribution is shown, and how solid the heatmap is. Every setting updates live, and the same colors are used on the whole-object map.



Vertical: share of the cut length per thickness, with lines at **Min** and **OK**.

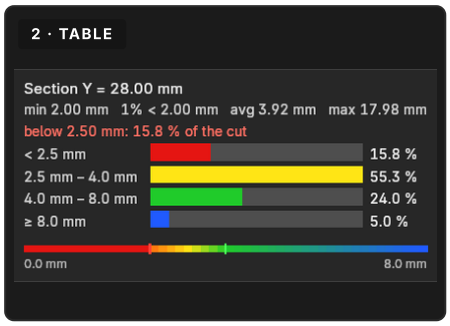
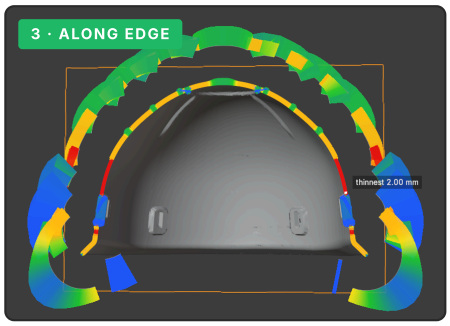


Table: one horizontal bar per class (below Min, Min–OK, OK–2×OK, thick) with the percentage. In Range mode: six bands.



Along Edge: the thickness drawn as a band along the outside of the cut, **Gap** 10 % of the cut size from the edge, **Height** = thickness.

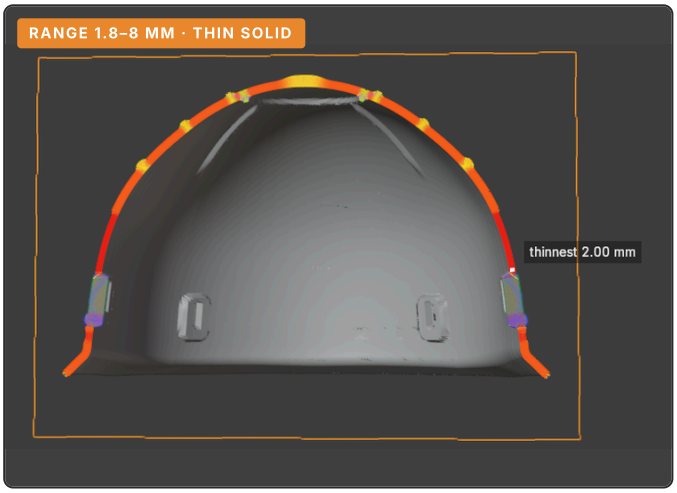
Colors: Min / OK or Range

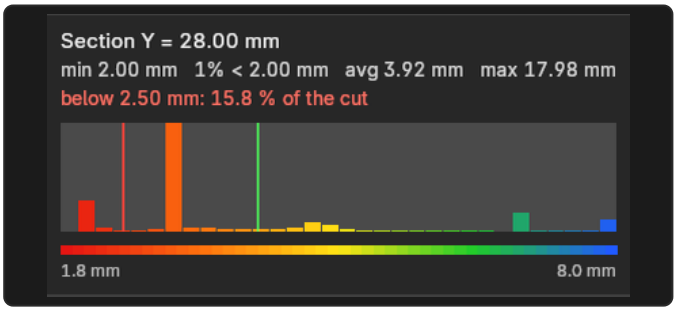
MODE	SCALE
Min / OK	red below Min, yellow up to OK, green to 2 × OK, blue above. Best to check a rule.
Range	continuous: **From** = red → orange → yellow → green → **To** = blue. Best to compare areas.
Auto Range	From / To from the data: 1 % to 99 % of the cut (or of the object analysis), rounded.

A strip with the scale and its end values sits under every histogram.

Opacity linked to thickness

OPACITY	RESULT
Solid	every wall fully colored.
Thin Solid	thin walls solid, thick walls fade: the problems stand out.
Thick Solid	the reverse: thick walls solid, thin ones fade.
Strength	0 = no fading, 1 = the faded end almost invisible.





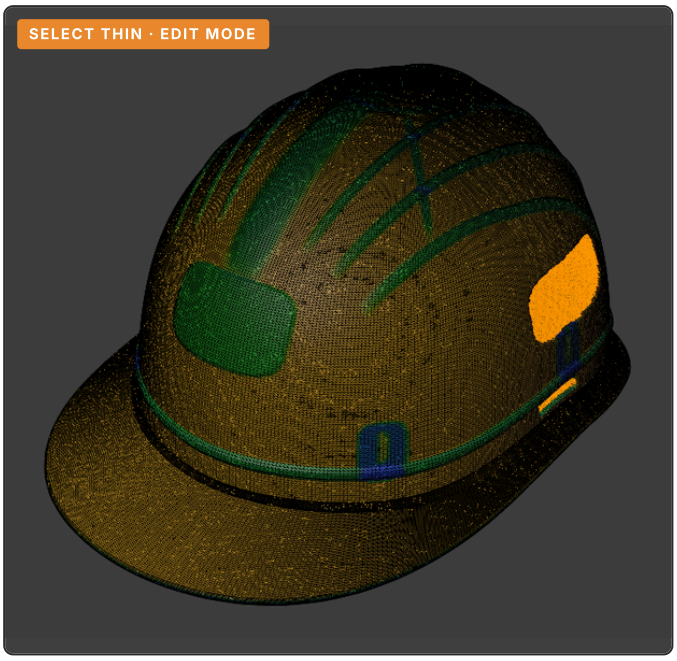
Auto Range set 1.8–8 mm on this cut. With **Thin Solid** the thin side walls stay solid red and the thicker crown fades.

06 WHOLE OBJECT

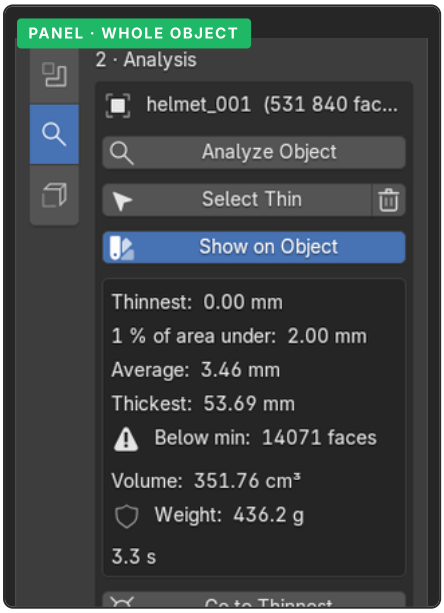
Every face measured.
The thin ones selected.

Analyze Object shoots one ray per face, inward along the face normal, to the opposite wall: the true wall thickness. It runs in the background in small slices with a progress bar, so Blender keeps responding; `Esc` cancels and leaves the mesh as it was. Then the object is colored with the same heat colors.





Left: 532k-face helmet colored by wall thickness. **Right: Select Thin** selects the 14,071 faces thinner than 2.5 mm in Edit Mode, so you see exactly where to add material.



What you get

- **Thinnest**, **1 % of area under**, **average** (area weighted), **thickest**, faces **below Min**.
- **Go to Thinnest**: 3D cursor inside the thinnest wall, view centered on it.
- **Show on Object**: viewport colors on / off. Change Min or OK and the colors follow, without measuring again.
- **Select Thin**: faces below Min selected in Edit Mode.
- **Live Map (Edit Mode)**: Blender's own thickness overlay, live while you add material.
- **Clear Results** removes the two stored attributes `mu_thickness` and `MU_Thickness`. Geometry is never changed.

Measured

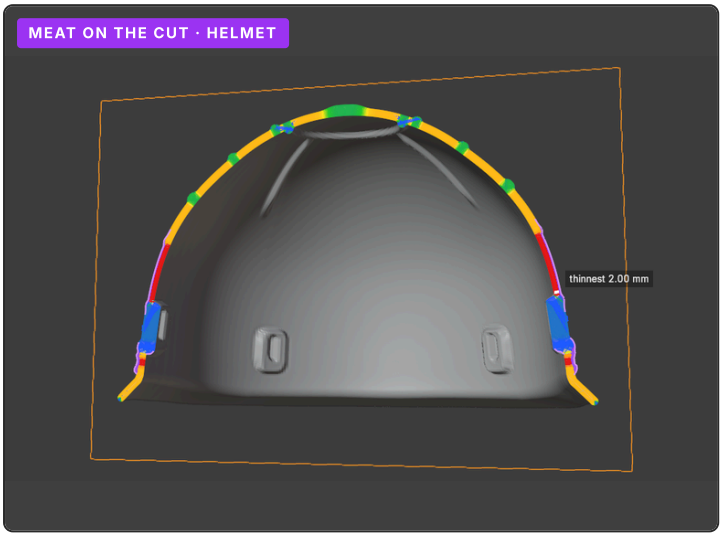
MESH	FACES	WHOLE OBJECT	SECTION MOVE
Helmet (helmet_001)	532k	3.3 s	< 50 ms
Dense raw scan (example)	2.9 M	22 s, in the background	≈ 50 ms

Windows laptop, Blender 5.2.3. The first move after Add Section Plane reads the mesh once (≈ 1.7 s for 2.9 M faces); later moves reuse it.

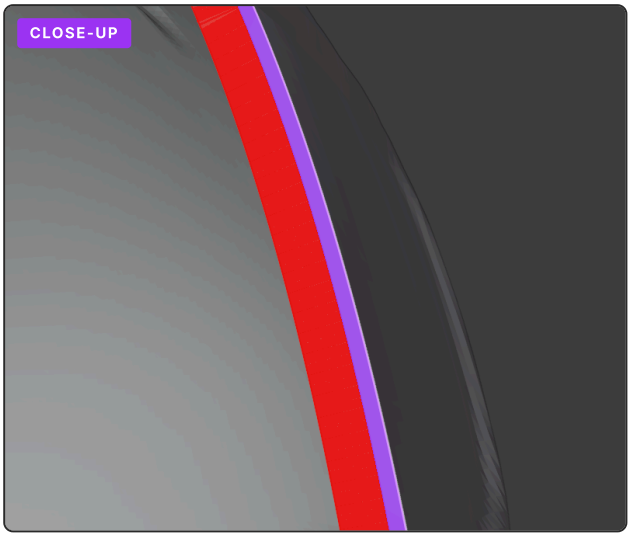
07 EXTRA · ADD MEAT

Extra: add the meat where the analysis found it thin.

The analysis is the core of MeshUnderskin; this is an optional next step. After **Analyze Object**, the **3 · Reinforce** tab makes a copy of the part and pushes its surface out only where the wall is thinner than **Min**, with a smooth blend into the rest. The original stays untouched (hidden), the added material shows on every cut in **purple**, and the panel tells you how many grams it costs.



Outside · To Min 2.5 mm + 0.2. The cut shows the wall before (red where too thin) and the added meat as a thin purple skin on the outside.



Close-up at the thinnest spot: red wall, purple meat on top.



Purple = where meat was added, grey = untouched. **Original / Thickened** switches the view.

Measured on the helmet · 532k faces, Min 2.5 mm

	BEFORE	AFTER
Faces below Min	14,071	227
1 % of area under	2.00 mm	2.70 mm
Weight, PLA	436.2 g	+6.3 g
Time	Add Meat 2.9 s · Check 3.8 s	

The 227 faces left are sharp rims and tight corners. 21 vertices were held back so the meat never touches a neighbouring wall.

1 · Choose

METHOD	WHAT YOU GET
Thicken	material added to thin walls: **Outside**, **Inside** or **Both**
Lines · Grid	parallel ribs · ribs in two directions (next page)
Honeycomb	hexagonal ribs (next page)

The material and weight are the ones you picked in **2 · Analysis**: no second choice here.

2 · Set · everything in mm, with sensible limits

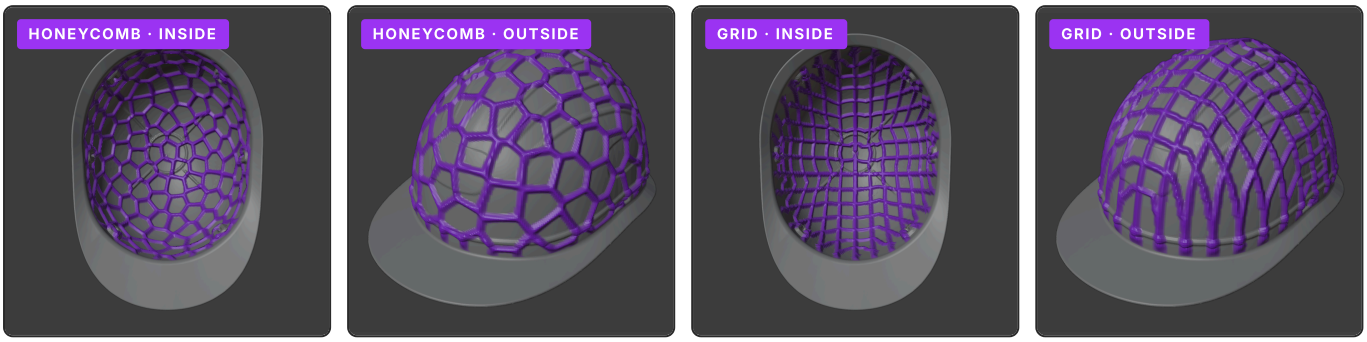
SETTING	RANGE
To Min + Margin	every thin wall up to Min + 0...5 mm (0.2)
Add mm	a fixed 0.1...20 mm where the wall is thin
Smooth edge	0...30 mm transition (3)
Max add	0...30 mm in one spot (0 = auto)

Then **3 · Generate**. Change anything and press **Regenerate**: it replaces the previous copy. **Light clean + smooth copy** (on by default) tidies the copy first: duplicate points merged, floating specks removed, scan noise smoothed very lightly without shrinking. The original is never touched.

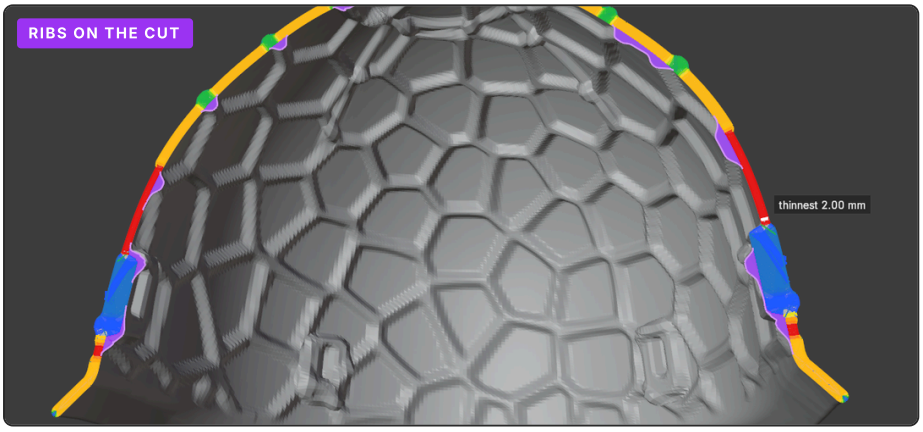
08 EXTRA · RIBS & HONEYCOMB

Stiffen it instead of filling it: ribs or honeycomb.

A rib adds stiffness for little weight: the bending stiffness of a wall grows with the cube of its height. Choose **Lines**, **Grid** or **Honeycomb**, set how many, how thick and how high (in mm), press **Generate**. Ribs go **Inside** or **Outside**, on the thin zones only or on the **Whole Object**.



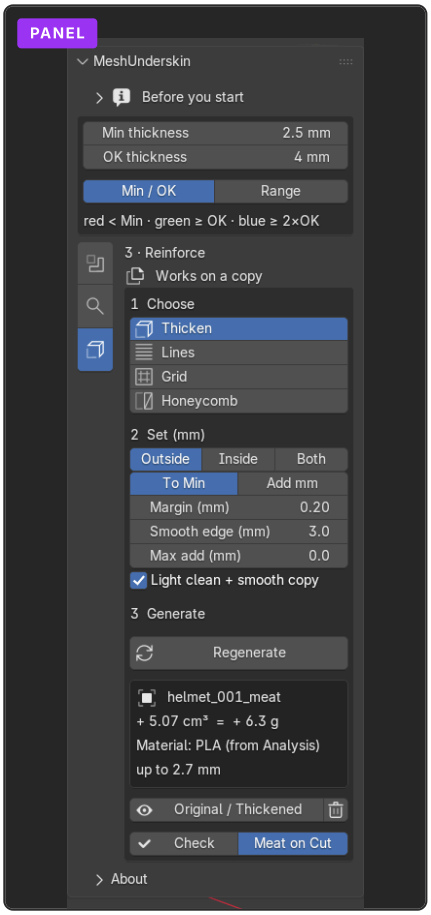
Same helmet, ribs drawn over the **whole helmet**, 14 cells across, ribs 4 mm thick and 3 mm high, copy cleaned and lightly smoothed first, **Edge fade** 8 mm: towards the rim the ribs die out smoothly instead of wrapping around onto the other side. The pattern is built on the surface itself, not projected: honeycomb cells grow from evenly spread points on the surface and are relaxed into near-regular hexagons; grid ribs are spaced by distance measured along the surface. The network is checked to be **one connected piece**: loose fragments are removed, so there are no cut "reinforcement bars". Added with these settings: honeycomb +91.3 g inside / +97.7 g outside, grid +102.9 g inside / +111.3 g outside (PLA), 5–8 s.



The section through the honeycomb copy shows every rib as a thicker (blue) bump on the inside of the wall, with the regular honeycomb behind it.

Rib settings

SETTING	MEANING
Auto	sets thickness, height, edge fade and smooth edge to suit the size of the object (done once when you open the tab). Ribs far too big for the object are refused before any work starts
Axis	Lines and Grid: the first rib runs across this axis, the others follow at equal distance along the surface. Honeycomb needs no axis
Ribs / Cells across	1…50: how many ribs (Lines) or cells across the part (Grid, Honeycomb). Honeycomb never exceeds **500 cells** in total
Thickness · Height	width of one rib 0.5…20 mm (2) · how far it stands off the wall 0.5…30 mm (3)
Edge fade	0…50 mm (8): ribs fade out smoothly towards the rim and never wrap onto the other side; 0 = off
Inside / Outside	which surface gets the ribs
Thin Zones / Whole Object	where ribs are made; counts are spread over that area
Also bring walls to Min	thicken the thin walls too, then put the ribs on top



The panel: choose, set in mm, generate. Here: Thicken outside, +5.07 cm³ = +6.3 g in PLA (the material from **2 · Analysis**) on a 436.2 g helmet. **Before you start** opens with a click.

Tip. A rib needs a few mesh edges across its width. If the ribs are narrower than two edges, MeshUnderskin warns you: make them wider or subdivide a copy first.

09 TIPS & LIMITS

Good to know before you trust a number.

“Cut face not shown”

Blender's **Manifold** solver cannot cut an open or non-manifold mesh, such as a raw scan that ends at the table. MeshUnderskin notices that the cut did nothing and says so. The heatmap and every number are still correct, because they are measured on the mesh itself. **Try Exact Solver** shows the cut face too, but it is slow on millions of faces.

Inside-out scans

If every normal of a scan points inwards, MeshUnderskin finds it from the signed volume and measures the right way anyway.

Grey on the cut

No opposite wall was found: an open edge of a scan, or a face lying flat in the cut plane. Move the plane a little.

Edited the mesh?

The add-on notices changes in vertex count and shape and measures again. After a big edit press **Re-fit to Object** so the plane fits the new size.

Workflow ideas

- **3D printing:** Min = your printer's smallest solid wall (e.g. 2 × nozzle), OK = the wall you design for.
- **Molding / casting:** walk the plane through the part in steps and watch for red and for blue next to each other (sink marks).
- **Reverse engineering:** read the wall of a scanned part at a few cuts and put the value in your CAD model.
- **Saving material:** blue areas are at least twice as thick as needed.
- Clean the scan first (holes, loose pieces) with MeshPharmacy: closed meshes cut better.

Never changes your mesh

The cut is a modifier, the plane and cutter are helper objects, the analysis writes two attributes you can clear, and Add Meat works on a copy. **Remove** (the X next to the object name) takes everything away. All buttons support Ctrl+Z.

Honest limits

A wall can only be measured where the mesh has it. Very thin double surfaces from scanning, self-intersections and sharp rims show up as tiny “thinnest” values: check them with **Go to Thinnest** and rely on **1 % under**. Treat the numbers as a visual guide, not a simulation, and check the real part.

Read this first: closed mesh, visual guide only

For correct section views and thickness maps the scan must be closed (no holes) and reasonably repaired. Open meshes, loose pieces and self-intersections give wrong or missing values. Repair first, for example with MeshPharmacy, then measure.

MeshUnderskin is not a simulation, strength or fatigue tool. Colours and numbers are an indicative guide to wall thickness. There is no load or material model. Always test the real part.

MeshUnderskin is part of an add-on family for scanning and reverse engineering, with MeshPharmacy (repair and clean scans) and ThreadPharmacy (rebuild threads). © 2026 Mirosław Pałucki · GPL-3.0-or-later.